

Surname(s) / APELLIDOS	Name / NOMBRE
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Exercise 1 [7 marks]

The following figures show a certificate chain and the contents of each certificate. This certificate chain has been returned when initiating a connection to www.upm.es. Based on the certificate chain displayed and the content of each certificate, answer the following questions:

Figure 1: Certificate Chain

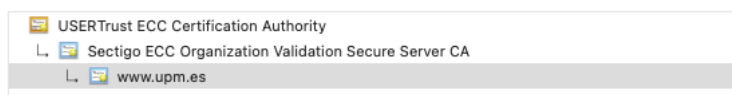


Figure 2: Certificates

USERTrust ECC Certification Authority	Sectigo ECC Organization Validation Secure Server CA	www.upm.es
Trust Root certificate authority Expires: Tuesday, 19 January 2038 at 00:59:59 Central European Standard Time This certificate is valid	Trust Intermediate certificate authority Expires: Wednesday, 1 January 2031 at 00:59:59 Central European Standard Time This certificate is valid	Trust Issued by: Sectigo ECC Organization Validation Secure Server CA Expires: Sunday, 9 June 2024 at 01:59:59 Central European Summer Time This certificate is valid
Details Subject Name Country or Region US County New Jersey Locality Jersey City Organisation The USERTRUST Network Common Name USERTrust ECC Certification Authority Issuer Name Country or Region US County New Jersey Locality Jersey City Organisation The USERTRUST Network Common Name USERTrust ECC Certification Authority Serial Number 5C 8B 99 C5 5A 94 C5 D2 71 56 DE CD 89 80 CC 26 Version 3 Signature Algorithm ECDSA Signature with SHA-384 (1.2.840.10045.4.3.3) Parameters None Not Valid Before Monday, 1 February 2010 at 01:00:00 Central European Standard Time Not Valid After Tuesday, 19 January 2038 at 00:59:59 Central European Standard Time Public Key Info Algorithm Elliptic Curve Public Key (1.2.840.10045.2.1) Parameters Elliptic Curve secp384r1 (1.3.132.0.34) Public Key 97 bytes: 04 1A AC 54 5A A9 F9 68 ... Key Size 384 bits Key Usage Verify	Details Subject Name Country or Region GB County Greater Manchester Locality Salford Organisation Sectigo Limited Common Name Sectigo ECC Organization Validation Secure Server CA Issuer Name Country or Region US County New Jersey Locality Jersey City Organisation The USERTRUST Network Common Name USERTrust ECC Certification Authority Serial Number 35 BE 74 63 8C B3 E9 DE 28 05 70 87 97 80 B3 67 Version 3 Signature Algorithm ECDSA Signature with SHA-384 (1.2.840.10045.4.3.3) Parameters None Not Valid Before Friday, 2 November 2018 at 01:00:00 Central European Standard Time Not Valid After Wednesday, 1 January 2031 at 00:59:59 Central European Standard Time Public Key Info Algorithm Elliptic Curve Public Key (1.2.840.10045.2.1) Parameters Elliptic Curve secp256r1 (1.2.840.10045.3.1.7) Public Key 65 bytes: 04 9C 8E 5C 0A 61 6F A1 ... Key Size 256 bits Key Usage Encrypt, Verify, Derive	Details Subject Name Country or Region ES County Madrid Organisation Universidad Politecnica de Madrid Common Name www.upm.es Issuer Name Country or Region GB County Greater Manchester Locality Salford Organisation Sectigo Limited Common Name Sectigo ECC Organization Validation Secure Server CA Serial Number 00 A1 0D CF A6 41 D6 17 4D 1A ED 6D 3D D4 9F 96 79 Version 3 Signature Algorithm ECDSA Signature with SHA-256 (1.2.840.10045.4.3.2) Parameters None Not Valid Before Friday, 9 June 2023 at 02:00:00 Central European Summer Time Not Valid After Sunday, 9 June 2024 at 01:59:59 Central European Summer Time Public Key Info Algorithm Elliptic Curve Public Key (1.2.840.10045.2.1) Parameters Elliptic Curve secp256r1 (1.2.840.10045.3.1.7) Public Key 65 bytes: 04 A2 6E 9F F7 33 0A F1 ... Key Size 256 bits Key Usage Encrypt, Verify, Derive Signature 70 bytes: 30 44 02 20 2B 6E 45 8F ...

- Write the common name of the domain returned by the web server certificate. [0,25 marks]
www.upm.es
- Write the serial number of that certificate. [0,25 marks]
00 A1 0D CF A6 41 D6 17 4D 1A ED 6D 3D D4 9F 96 79
- Which company issued that certificate? [0,25 marks]
Sectigo Limited - Sectigo ECC Organization Validation Secure Server CA
- What kind of public key is included in the webserver certificate? [0,25 marks]
It includes a public key for an elliptic curve in which the curve and the 256-bit public key are defined
- Explain briefly the role of the public key included in the webserver certificate. [1,5 marks]

That public key is used to establish a session key that will be used to encrypt the connection to the server using, normally, a symmetric AES-type cipher.

- f) What algorithm was used to sign the webserver certificate? [0,5 marks]

It was signed with ECDSA and SHA-256

- g) Write down the size of the public key used to sign the webserver certificate. Write also the first 4 bytes of that key (you can write them in hexadecimal). [0,25 marks]

Signing requires the use of the private key. Another valid interpretation is that the key was 256 bits long and it starts with 04 9C.

- h) Write the name and common name of the root certification authority. [0,25 marks]

The USERTRUST Network - USERTrust ECC Certification Authority

- i) When does the CA certificate expire? [0,25 marks]

19th January 2038

- j) Write down the size of the CA public key. Write also the first 4 bytes of that key (you can write them in hexadecimal). [0,25 marks]

The key has 384 bits starting with 04 1A

- k) Explain why this certificate has a different key length than the other certificates [0,5 marks]

This certificate expires much later and also is used to sign all the other certificates so its key must sustain the improvements in computing speed that will happen while the certificate is valid.

- l) What certificates has the CA public key signed? [1 mark]

Two, the own CA certificate and the intermediate one. Another valid interpretation is that the signature requires the private key so no certificates were signed with the CA public key

- m) Briefly explain why the browser trusts the certificate [1,5 mark]

The browser trusts the certificate as it comes pre-installed on it. In order to pre-install the certificate in the browser, the CA has had to go through a series of controls and audits that ensure that the security processes carried out in it and the identity verification processes meet minimum standards.

Exercise 2 [3 marks]

Bob has sent Alice a document with his corresponding signature. To calculate the signature, Bob used RSA and a hash function that consists of dividing the message into 2-byte blocks and performing a sum of all the blocks modulo (2^9-1) . With the document data provided, the corresponding signature, and Alice and Bob public keys, answer the following questions:

- Document (in hexadecimal): [A73B 6C41]
- Signature (in hexadecimal): [0x29]
- Bob's public key (both in decimal): {e=13 , n=55}
- Alice's public key (both in decimal): {e=13 , n=77}

- a) Explain why this particular hash function is a bad choice for the signature algorithm [1 mark]

This is a very poor hash function that is not resistant to pre-image or second image attacks. Therefore it is very easy to find collisions so this means that more than one document will generate the same signature and therefore Bob will be able to deny that he signed the document he sent.

b) Verify the signature sent by Bob from Alice's point of view. [2 marks]

The parameters that have been used to perform the calculations are as follows.

For RSA. $P=5$ $q=11$. $N=55$ y $\phi(n)=40$. $E=13$ y $d=37$

To verify the signature, we first calculate the hash of the document. To do this, we convert the numbers to decimal, add them and reduce them mod 2^9-1

$0xA73B = 42811$

$0x6C41 = 27713$

$(0xA73B + 0x6C41) \bmod 2^9-1 = 6$ (in decimal)

That's the hash value. To verify the signature, what we do is decrypt the $0x29$ which is 41 in decimal (using Bob's public key).

$41^{13} \bmod 55 = 6$ so the signature is correct